PROJECT REPORT

ON

THE STUDY OF OPERATIONS EXPOSURE AT AV ENGINEERS PVT. LTD.

SUBMITTED IN THE FULFILMENT OF THE REQUIREMENTS FOR THE DEGREE OF

MASTER OF BUSINESS ADMINISTRATION

Affiliated to Himachal Pradesh Technical University Hamirpur
(SESSION: 2024-2026)



SUBMITTED TO:
Department of MBA

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DECLARATION

I am Priya, a student of **GOVT. PG College Dharamshala**, affiliated to **Himachal Pradesh Technical University, Hamirpur – 177001, Himachal Pradesh**, hereby declare that I have completed my project report on **"Study Of Operation Exposure in AV Engineers Pvt. Ltd."** in partial fulfilment for the award of the Degree in **Master of Business Administration** of Session **(2024–26)**. The information submitted is true and original to the best of my knowledge.

(Signature of Student)

Priya

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

(Signature of supervisor)

Dr.Meghna Sood
(Faculty,Govt. Pg College Dhramshala)

Date: _____

The project report viva voce examination of Priya has been held on……..

Expert outside of Department Expert from the Department Supervisor

CERTIFICATE FROM COMPANY

CERTIFICATE FROM GUIDE

This is to certify that the project report entitled " **Study Of Operation Exposure in AV Engineers Pvt. Ltd.** "has been completed under my supervision and guidance. It was submitted in partial fulfilment of the requirements for the award of the degree of MASTER OF BUSINESS ADMINISTRATION from **GOVT. PG College Dharamshala**. The project was carried out by Miss Priya, bearing Roll Number 240628010042, with dedication and commitment. Throughout the project duration, **Miss Priya** demonstrated a strong work ethic, research skills, and analytical thinking. The findings and conclusions presented in the project report are the outcome of his diligent efforts and academic abilities.

I have closely supervised and provided guidance to **Miss Priya** during the various stages of the project. I have reviewed the report's content, methodology, and overall quality, ensuring that it meets the necessary academic standards. Therefore, based on my assessment, I certify that the project report has been successfully completed under my supervision. It is my belief that **Miss Priya** has met the academic requirements set forth by Himachal Pradesh Technical University for the degree of MASTER OF BUSINESS ADMINISTRATION.

I wish **Miss Priya** continued success in his academic and professional endeavours. Sincerely,

Dr. Meghna Sood
Project Supervisor

ACKNOWLEDGEMENT

Perseverance, In the present world of competition there is a race of existing in which those who are willing to come forward succeed. The project is like a bridge between theoretical and practical work. I joined this project. To design and compare a project report is very laborious work, which no student completes without taking any help from any professional.

First of all, I would like to thank the supreme power of almighty God who is obviously the one who has always guided us to work in the right path of our life.

I express my deep gratitude to my guide **Dr. Meghna Sood** for his invaluable guidance during the project. His unlimited guidance, innovative ideas and tireless efforts helped along the way in completing the project. I am also thankful to the staff members for their encouragement and cooperation in this successful completion of my project. My special thanks to my classmates and dear friends, for their support throughout my internship. With their help I could complete my work efficiently and effectively.

Last but not the least, my endless appreciation goes to my family who has stood by my side and given me moral support whenever I was low and boosted my will power. In the end I would like to thank my parents who greatly indebted me for having brought me love and encouragement at this stage.

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EXECUTIVE SUMMARY

This project report focuses on the study of Operations Exposure at AV ENGINEERS PVT. LTD., Chandigarh. The main purpose of the study is to understand how operational activities, business processes, market conditions, and external environmental factors affect the overall performance and profitability of the company. The study highlights the importance of identifying operational risks and managing them effectively to improve efficiency, productivity, and decision-making.The project provides an overview of the company's operations, organizational structure, and business environment. It also explains the concept of operations exposure and its impact on production, supply chain management, cost control, inventory management, and customer satisfaction. The report examines the challenges faced by the company due to fluctuations in raw material prices, market competition, technological changes, and operational inefficiencies.

For conducting the study, primary data sources were used. Primary data was collected through questionnaires and interactions with employees and management, while secondary data was collected from company records, reports, websites, and other published sources.

The findings of the study reveal that effective operational planning and risk management play a significant role in maintaining organizational stability and improving performance. The study also indicates that proper coordination between departments, efficient inventory control, timely decision-making, and the use of modern technology can reduce operational exposure and increase productivity.

Based on the analysis, the report suggests that AV ENGINEERS PVT. LTD. should strengthen its operational strategies, improve communication systems, adopt advanced technologies, and enhance employee training programs. These measures can help the company minimize operational risks, improve efficiency, and achieve long-term growth and sustainability.

Overall, the project concludes that operations exposure management is essential for ensuring smooth business functioning and maintaining a competitive position in the market. The study provides useful insights and practical recommendations that can support the company in improving its operational effectiveness and future business performance.

CHAPTER I
INDUSTRY PROFILE

Operations Exposure at AV Engineers Pvt. Ltd., Chandigarh

1.1 Introduction of the Industry

The Heating, Ventilation, Air Conditioning and Refrigeration (HVAC&R) industry plays a vital role in modern infrastructure and industrial development. This industry is responsible for maintaining thermal comfort, regulating indoor air quality, and ensuring proper ventilation in enclosed spaces.HVAC systems are widely used in residential buildings, commercial complexes, industries, and institutional setups. Refrigeration systems, on the other hand, are essential for preserving perishable goods, medicines, and chemicals by maintaining controlled temperature conditions.In earlier times, HVAC systems were considered luxury items, but with rising temperatures, urbanization, and changing lifestyles, they have now become a necessity. The increasing demand for comfort, health, and productivity has further boosted the adoption of HVAC systems.The HVAC&R industry is also closely linked with energy consumption, as these systems require electricity to operate. Therefore, modern developments in the industry are focused on energy efficiency, sustainability, and environmentally friendly refrigerants.Companies like AV Engineers Pvt. Ltd. operate within this industry by providing end-to-end HVAC solutions, including system design, installation, testing, and maintenance services.

1.2 Overview of HVAC&R Industry in India

The HVAC&R industry in India has been growing rapidly over the past decade and is expected to continue its growth in the coming years. Several factors contribute to this expansion:

a) Urbanization and Smart Cities

Rapid urban development and government initiatives like Smart Cities Mission have increased the construction of modern buildings requiring advanced HVAC systems.

b) Rising Income and Lifestyle Changes

With increasing disposable income, people are investing more in comfort-based products such as air conditioners and ventilation systems.

c) Industrial and Commercial Growth

Industries such as pharmaceuticals, IT, manufacturing, and food processing require temperature-controlled environments, which drives demand for HVAC systems.

d) Climate Conditions

India's hot and humid climate significantly increases the need for cooling systems, especially during summer seasons.

e) Energy Efficiency and Sustainability

There is growing awareness regarding energy conservation, leading to the adoption of energy-efficient HVAC technologies like inverter ACs, VRV/VRF systems, and green building solutions.

f) Growth of Cold Chain Industry

The expansion of food delivery services, supermarkets, and pharmaceutical storage has increased the demand for refrigeration systems.

1.3 Nature of Industry Operations

The HVAC&R industry operates through a systematic and technical process involving multiple stages:

a) System Design and Engineering

This is the initial stage where engineers analyze building requirements, cooling load calculations, and environmental factors to design suitable HVAC systems.

b) Procurement of Equipment

This involves purchasing essential components such as compressors, condensers, evaporators, ducts, pipes, insulation materials, and control systems from vendors.

c) Installation and Execution

Trained technicians and engineers install HVAC systems at project sites. This includes ducting work, piping, electrical connections, and system setup.

d) Testing and Commissioning

After installation, systems are tested to ensure proper functioning, efficiency, and safety before being handed over to the client.

e) Maintenance and After-Sales Service

Regular servicing is required to maintain system efficiency. Companies provide AMC (Annual Maintenance Contracts), repair services, and system upgrades.

1.4 Industry Applications

The HVAC&R industry serves a wide range of sectors, including:

- Residential buildings (apartments, villas)
- Commercial complexes (malls, offices)
- Hospitality sector (hotels, restaurants)
- Healthcare (hospitals, laboratories)
- Education (schools, colleges)
- Industrial units (factories, manufacturing plants)
- Pharmaceuticals (temperature-sensitive production areas)
- Cold storage and warehouses
- IT sector and data centers

Each sector has different requirements, making the industry highly specialized and diversified.

1.5 Operations Management in HVAC Industry

Operations management is a crucial aspect of the HVAC&R industry as it ensures efficient project execution and customer satisfaction.

a) Project Planning and Scheduling

Proper planning is required to allocate resources, set timelines, and manage project execution efficiently.

b) Resource Management

This includes managing manpower, machinery, and materials to ensure smooth workflow.

c) Inventory and Supply Chain Management

Maintaining an adequate stock of equipment and coordinating with suppliers is essential to avoid delays.

d) Quality Control and Safety

Strict quality standards must be followed during installation and maintenance to ensure system reliability and safety.

e) Cost and Budget Control

Monitoring project costs, minimizing wastage, and ensuring profitability are key responsibilities of operations management.

f) Coordination and Communication

Operations require coordination between engineers, project managers, suppliers, and clients to ensure successful project completion.

1.6 Challenges in HVAC&R Industry

Although the HVAC&R (Heating, Ventilation, Air Conditioning, and Refrigeration) industry is growing rapidly, it also faces several practical challenges that affect its operations and performance.

a) High Initial Investment

One of the major challenges is the high cost of installing HVAC systems. Advanced equipment, design work, and installation require a significant investment. This can sometimes discourage customers, especially in small or budget-sensitive projects, from adopting high-quality systems.

b) Technological Advancements

The HVAC industry is constantly evolving with new technologies and innovations. While this brings improvements, it also creates a challenge for companies and workers to keep up. Continuous learning, training, and upgrading of skills are required, which can be time-consuming and costly.

c) Energy Consumption Issues

HVAC systems consume a large amount of electricity, especially in commercial and industrial setups. This leads to higher operating costs for customers. As a result, there is increasing pressure on companies to develop and provide more energy-efficient solution

d) Skilled Labor Shortage

There is a shortage of skilled technicians and trained professionals in the HVAC industry. Since the work involves both technical knowledge and practical skills, finding experienced workers can be difficult. This can affect the quality of work and project timelines.

e) Environmental Regulations

The government has strict rules regarding the use of refrigerants and energy efficiency to protect the environment. Companies must follow these regulations carefully, which sometimes increases costs and requires changes in technology and processes.

f) Project Delays and Cost Overruns

HVAC projects often depend on multiple factors such as material availability, site readiness, and coordination between teams. Any delay in these areas can lead to increased costs and missed deadlines, affecting both the company and the client.

1.7 Conclusion of industry profile

The industry in which AV Engineers Pvt Ltd operates plays a vital role in supporting industrial growth, infrastructure development, and commercial operations. The HVAC and engineering solutions industry has witnessed continuous growth due to increasing industrialization, urbanization, rising demand for energy-efficient systems, and technological advancements in manufacturing and cooling solutions.
The industry provides essential services related to heating, ventilation, air conditioning, refrigeration, industrial automation, and engineering support to various sectors such as manufacturing, pharmaceuticals, food processing, retail, hospitality, healthcare, and commercial infrastructure. With increasing competition and globalization, companies are focusing on operational efficiency, cost reduction, quality improvement, and customer satisfaction to maintain their market position.
In recent years, the industry has also experienced significant changes because of advancements in technology, automation, digital monitoring systems, and environmentally sustainable practices. The adoption of energy-efficient equipment and eco-friendly refrigerants has become important due to government regulations and environmental concerns. This has created new opportunities for companies to innovate and improve their operational performance.

Chapter II

COMPANY PROFILE

2.1 Introduction

AV Engineers Pvt. Ltd. is a well-established and reputed company in the field of HVAC (Heating, Ventilation, Air Conditioning & Refrigeration) systems. The company has built a strong presence across North India by delivering advanced, reliable, and energy-efficient HVAC solutions to a wide range of industries and commercial sectors. With years of experience and technical expertise, AV Engineers specializes in providing complete turnkey solutions, including system design, installation, commissioning, maintenance, and after-sales support.

The company is committed to maintaining high standards of quality, innovation, and customer satisfaction while ensuring cost-effective and environmentally friendly engineering solutions. AV Engineers serve various sectors such as commercial buildings, hotels, hospitals, educational institutions, industrial facilities, restaurants, and retail spaces. Backed by a skilled team of professionals and modern technology, the company continuously strives to deliver efficient climate-control solutions that meet the evolving needs of its clients.

2.2 Company Overview

- Founded in the mid-1990s (around 1996–1998) by experienced technocrats
- Headquartered in Chandigarh, India
- Has more than 25+ years of industry experience in HVAC and refrigeration projects

- The company started with the vision of combining modern technology with business needs and has grown into a trusted HVAC solution provider.

- Registered office - Head Office:Sector 45C, Kesho Ram Complex, Chandigarh – 160047, (India)

Basic Company Information

Particulars	Details
Company Name	A V Engineers Pvt. Ltd.
Industry	HVAC/R Engineering and Refrigeration
Headquarters	Chandigarh, India
Founded	1996
Business Type	Private Limited Company
Core Services	HVAC, VRV/VRF Systems, Refrigeration, Cold Rooms, Ventilation
Managing Director	Mr. Amit Munjal
Operational Area	North India and nearby regions

2.2.1 Nature of Business

AV Engineers Pvt. Ltd. operates in multiple roles within the HVAC industry, which makes it a complete solution provider for clients.

The company works as an **HVAC system contractor**, meaning it takes responsibility for designing and installing heating, ventilation, and air conditioning systems. It also acts as an **equipment supplier and distributor**, providing essential components like compressors, ducts, and other HVAC materials.

In addition, the company functions as a **project execution firm**, where it manages projects from start to finish. It also provides **service and maintenance**, ensuring that installed systems continue to work efficiently over time.

The company undertakes **turnkey projects**, which means it handles everything from beginning to end. This includes:

- **Design** – understanding client requirements and planning the system
- **Supply** – arranging all necessary equipment and materials
- **Installation** – setting up the HVAC system at the site
- **Testing** – checking system performance
- **Commissioning** – making the system fully operational
- **After-sales service** – providing maintenance and support

2.2.2 Core Services

- **AV Engineers Pvt. Ltd.** offers a wide range of services in the HVAC and engineering domain. The company mainly specializes in providing complete cooling and refrigeration solutions based on client requirements.
- Its key services include **HVAC and refrigeration solutions**, where it designs and installs systems for different types of buildings and industries. It also provides advanced systems like **VRV/VRF air conditioning**, which are known for energy efficiency and flexibility, along with **central air conditioning systems** for large spaces.
- In the area of industrial applications, the company handles **industrial refrigeration**, **cold storage**, and **cold rooms**, which are essential for preserving goods like food and medicines. It also works with specialized systems such as **ammonia and brine systems**, used in large-scale industrial cooling.
- Apart from installation, the company offers **engineering services** like system design and planning, ensuring that each project is customized and efficient. It also manages **project**

execution, taking responsibility from start to finish, and provides **maintenance and AMC (Annual Maintenance Contract) services** to keep systems running smoothly.

- A strong focus is also placed on **energy-efficient solutions**, helping clients reduce energy consumption and operational costs.

2.2.3 TARGET SEGMENT

- The services of **AV Engineers Pvt. Ltd.** are used across multiple sectors, as HVAC systems are essential in almost every type of building.
- The company works with **hotels and restaurants**, where proper cooling and ventilation are important for customer comfort. It also serves **hospitals and laboratories**, where temperature control is critical for safety and hygiene.
- In the commercial sector, it provides solutions for **offices and banks**, ensuring a comfortable working environment. It also undertakes projects for **educational institutions**, such as schools and colleges.
- Additionally, the company supports **industrial plants**, where specialized cooling systems are required for machinery and processes. It also works on **residential buildings**, providing air conditioning solutions for homes and apartments.

Table 2.1-

1. HOTELS & RESTAURANTS

Sr. No.	PROJECT NAME	LOCATION	HP	APPLICATION
1	THE BREW ESTATE	PATIALA	40	BREWERIES
2	COFFEE BEAN & TEA LEAF	MOHALI	16	COFFEE SHOP
3	COFFEE BEAN & TEA LEAF	ELANTE	11	COFFEE SHOP
4	PETER RABBIT	MOHALI	12	COFFEE SHOP
5	NAMASTE CHAAYOS	DERABASSI	10	TEA HOUSE
6	NUGGET NATION	MOHALI	8	TAKE AWAY
7	CPWD	PARAGPUR	80	GUEST HOUSE
8	IIT MANDI	MANDI	124	GUEST HOUSE
9	PUNJAB STATE POWER COPORATION	BATHINDA	50	GUEST HOUSE
10	EDDIES BISTRO	CHANDIGARH	36	RESTAURANT
11	GOPAL SWEETS	PATIALA	20	RESTAURANT
12	GOPAL SWEETS	AMBALA	66	RESTAURANT
13	GOPAL SWEETS SEC 35, CHANDIGARH	CHANDIGARH	21	RESTAURANT
14	GOPAL SWEETS SEC-70	MOHALI	26	RESTAURANT
15	KAMATS	CHANDIGARH	16	RESTAURANT
16	AMOHA DHARAMSHALA	DHARAMSHALA	190	HOTEL
17	BAJAJ HOTEL, SEC 66 A , MOHALI	MOHALI	30	HOTEL
18	COUNTRY INN	ZIRAKPUR	460	HOTEL
19	EMERALD	CHANDIGARH	30	HOTEL
20	HOTEL M-1	JALANDHAR	216	HOTEL
21	HOTEL ROYAL TULIP PVT. LTD	KUFRI,SHIMLA	36	HOTEL
22	HOTEL , KAMLANAGR, SHIMLA	SHIMLA	92	HOTEL
23	HOTEL ICON	CHANDIGARH	102	HOTEL
24	HOTEL MANDRIN	ZIRAKPUR	78	HOTEL
25	HOTEL MANDRIN-1	ZIRAKPUR	55	HOTEL
26	HOTEL MANDRIN-2	ZIRAKPUR	65	HOTEL
27	HOTEL RADISSON	JALANDHAR	420	HOTEL
28	HOTEL THE GOLDEN PALMS & SPA (STARWOOD HOTEL & RESORT)	SOLAN	120	HOTEL
29	HOTELS HERITAGE	CHANDIGARH	28	HOTEL
30	LEMON TREE	HISSAR	494	HOTEL
31	MAYA HOTEL	CHANDIGARH	144	HOTEL
32	NAYYAR ESTATE	MOHALI	34	HOTEL
33	NORBU, THE MONTANA, BANQUET HALL	DHARAMSHAALA	24	HOTEL
34	RADISSON HOTEL	KUFRI	30	HOTEL
35	RAMADA HOTEL	AMBALA	400	HOTEL
36	RATHORE HOTEL	SHIMLA	30	HOTEL
37	SAROVAR PORTICO	JALANDHAR	262	HOTEL
38	SHOGHI HOTEL	SHIMLA	48	HOTEL
39	SURYA VILAS LUXARY RESORT & SPA	KUMAR HATTI	208	HOTEL
40	TIMBER TRAIL	DHARAMSHALA	112	HOTEL
41	VILLAS NO 18, SANAWAR HILLS	HP	16	HOTEL
42	WINDSOR	KASAULI	48	HOTEL
43	WINNIES HOLIDAYINN	KASAULI	40	HOTEL

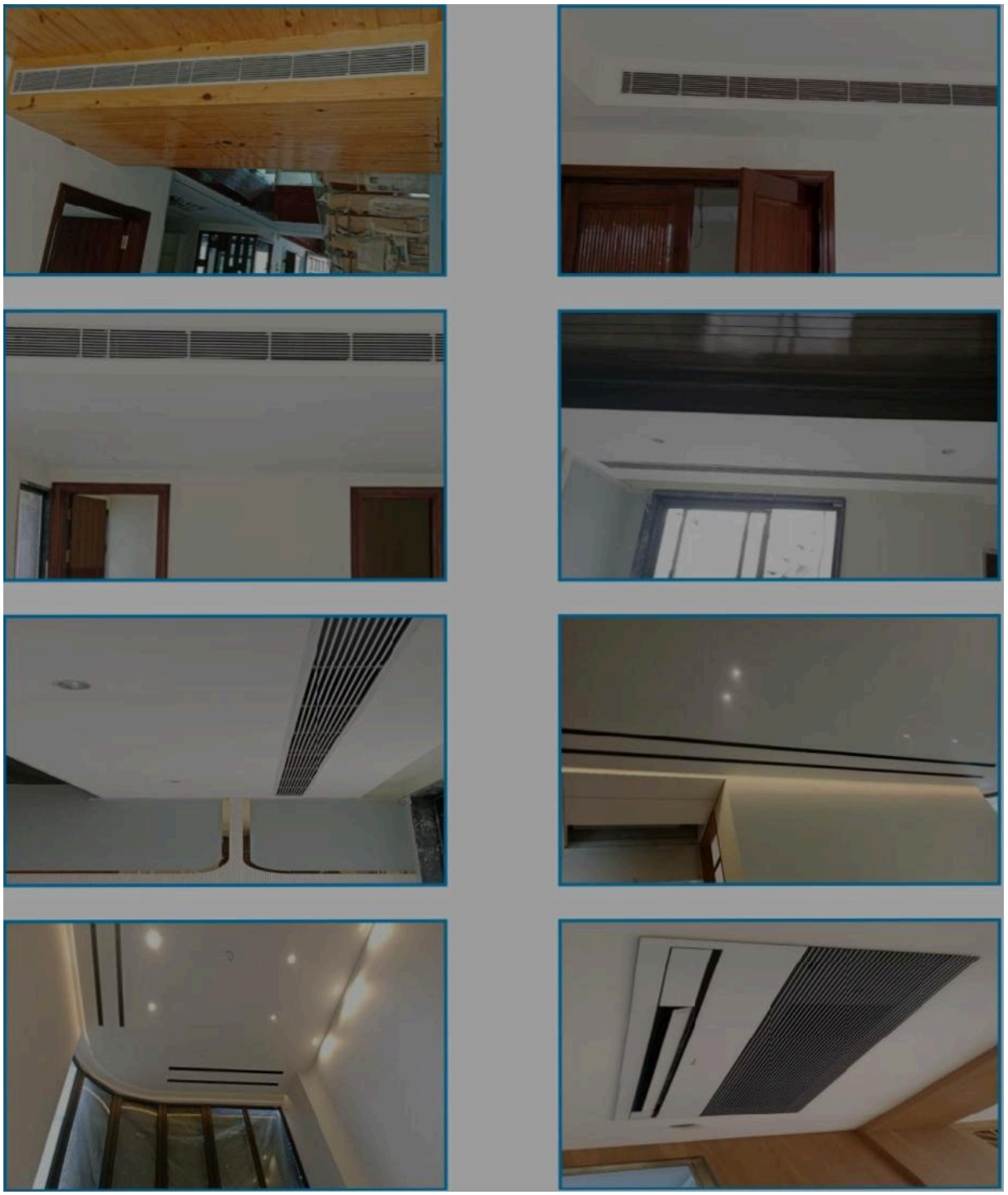
Table 2.2-

2. HOSPITALS & LABS

Sr. No.	Project Name	Location	HP	Application
01	DRDO LAB	RAMGARH	352	LAB
02	AMAR HOSPITAL	PATIALA	256	HOSPITAL
03	ARMAAN HOSPITAL	JALANDHAR	190	HOSPITAL
04	CARDINOVA HOSPITAL	JALANDHAR	150	HOSPITAL
05	ORTHONOVA HOSPITAL	JALANDHAR	108	HOSPITAL
06	KIDNEY HOSPITAL	JALANDHAR	54	HOSPITAL
07	DHAWAN HOSPITAL	PANCHKULA	34	HOSPITAL
08	SILVER OAKS HOSPITAL	CHANDIGARH	28	HOSPITAL
09	PET SCAN	CHANDIGARH	24	CLINIC
10	DR. GOURAV JAIN	CHANDIGARH	24	CLINIC
11	DR. BANSAL CLINIC	MANIMAJARA	16	CLINIC
12	KAMAL NURSING HOME	ROPAR	54	NURSING HOME

REAL INSTALLATIONS PICTURES

Fig No. 2.1



SCHOOLS & COLLEGES

Fig No. 2.2



IN-HOUSE TEAM

Fig No. 2.3

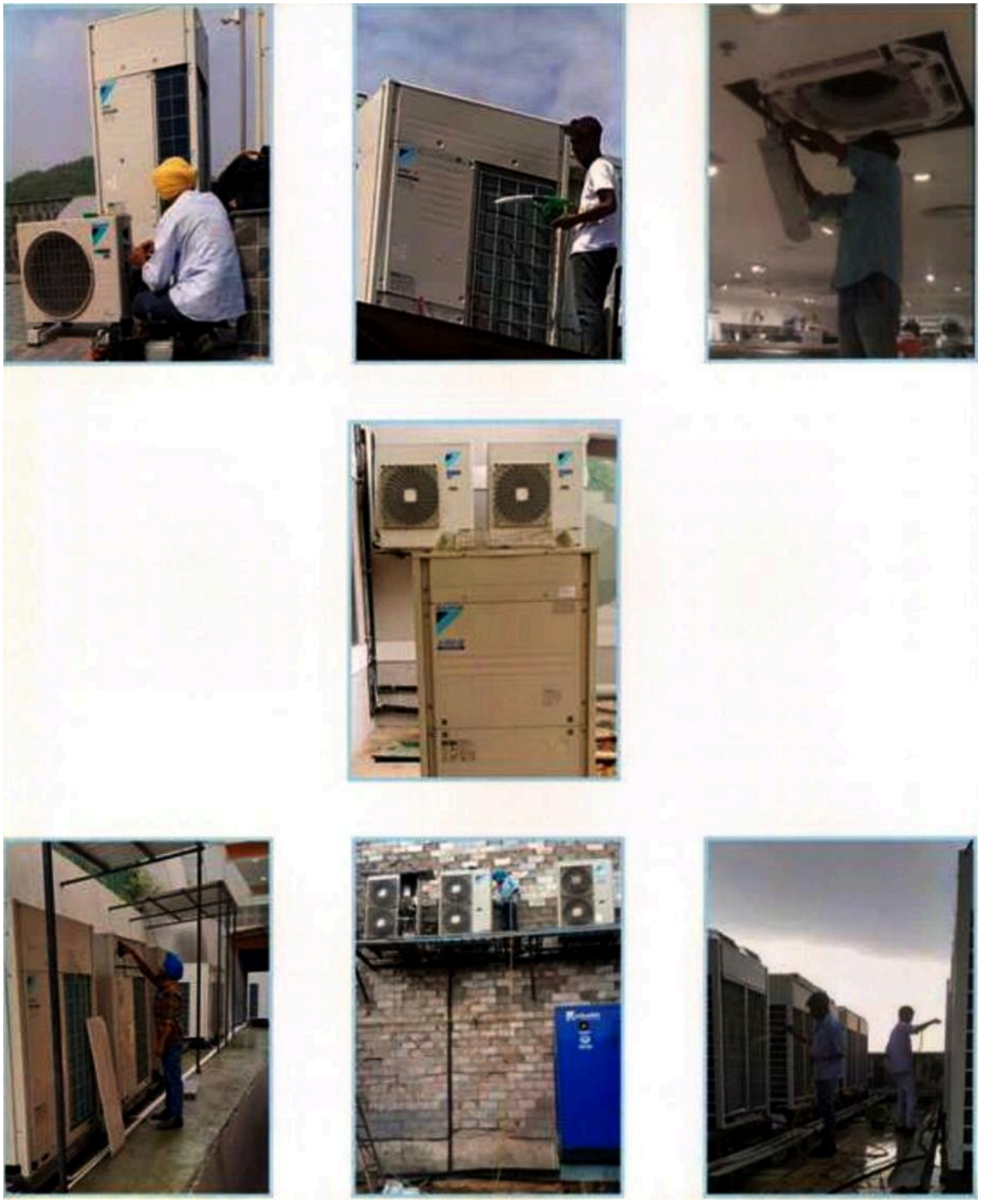


Fig No. 2.4


PARTNERS IN BUSINESS
DAIKIN
Danfoss
MEXFLOW
COPPER PIPES
Quality & Dependability
India ki Life Line
uponor
GRUNDFOS
INNOVATIVE

2.3 VISION & MISSION

The vision of AV Engineers Pvt. Ltd. is-

The company wants to become a trusted leader in the HVAC-R industry by using advanced technology, excellent service and innovative cooling solution

The company vision emphasizes -

- Blending modern technology which current business challenges
- Becoming a leading HVAC-R solutions provider in north india
- Building long-term customer relationship through quality and trust

The mission of AV engineers Pvt. Ltd. is-

To make available quality and reliable HVAC/R solutions, at optimal costs to all market segments while ensuring the highest standards of ethics and business practices at all steps of the value chain.

The company aims to achieve this mission by:

- Providing high-quality and customized HVAC/R solutions.
- Expanding air-conditioning services to different market segments.
- Delivering cost-effective cooling systems.
- Offering strong after-sales service support.
- Ensuring timely project execution and customer satisfaction.
- Maintaining ethical business practices and professional standards.

2.4 SWOT analysis

Strengths-

- **Strong Industry Experience**

The company has more than 25 years of experience in HVAC/R systems, which provides employees and trainees with practical exposure to real industrial operations and project execution.

- **Wide Range of Operations**

AV Engineers works in HVAC design, installation, testing, commissioning, refrigeration systems, VRV projects, and after-sales service. This gives exposure to multiple operational functions within the engineering industry.

- **Turnkey Project Handling**

The company manages complete turnkey projects, helping students understand planning, procurement, execution, maintenance, and customer service in real business conditions.

- **Experienced Technical Team**

The organization has skilled engineers, technicians, project managers, and service staff, which supports learning and operational training.

- **Regional Market Presence**

AV Engineers operates across North India including Punjab, Haryana, Himachal Pradesh, Delhi, and nearby regions, giving exposure to diverse projects and customer requirements.

- **Association with Leading Brands**

The company is an authorized dealer of Daikin Airconditioning India Pvt. Ltd., which increases technical credibility and learning opportunities in advanced HVAC technologies.

Weaknesses-

- **Limited International Exposure**

Most operations are concentrated in North India, so international operational exposure may be limited compared to multinational HVAC companies.

- **Dependence on Construction and Infrastructure Sector**

Business performance is highly dependent on real estate, infrastructure, and industrial projects. Slowdowns in these sectors can affect operations.

- **High Operational Costs**

HVAC projects require expensive machinery, skilled manpower, transportation, and maintenance, which increases operational expenses.

- **Workforce Management Challenges**

Managing technicians, engineers, and field staff across multiple project locations can create coordination and communication difficulties.

- **Technology Upgradation Requirement**

Continuous investment is required to stay updated with energy-efficient and smart HVAC technologies.

Opportunities -

- **Growing Demand for HVAC Systems**

Increasing urbanization, commercial buildings, hospitals, malls, hotels, and smart infrastructure projects are increasing the demand for HVAC services in India.

- **Expansion into Green and Energy-Efficient Solutions**

Rising awareness about energy conservation provides opportunities in eco-friendly HVAC systems and sustainable cooling technologies.

- **Digital and Smart Building Technologies**

Integration of automation, IoT, and smart temperature control systems can improve operational efficiency and create new business opportunities.

- **Skill Development and Training**

Students and trainees can gain strong practical knowledge in operations management, project coordination, technical maintenance, and customer handling.

- **Expansion into New Markets**

The company can expand operations to other states and international markets for larger business growth.

Threats-

- **Intense Competition**

The HVAC industry has strong competition from national and multinational companies, which can affect market share and profitability.

- **Rapid Technological Changes**

Frequent technological advancements require continuous employee training and investment in modern equipment.

- **Economic Slowdowns**

Recession or decline in construction activities can reduce demand for HVAC installation projects.

- **Fluctuating Material Costs**

Prices of compressors, refrigerants, copper pipes, and other HVAC components can increase operational costs.

- **Government Regulations**

Environmental laws regarding refrigerants and energy efficiency standards may increase compliance requirements.

- **Skilled Labor Shortage**

Difficulty in retaining trained technicians and engineers may affect project execution efficiency.

2.5 CONCLUSION OF COMPANY PROFILE

AV Engineers Pvt. Ltd. also provides valuable exposure to modern HVAC technologies, project management practices, operational coordination, and service management. The company continuously focuses on innovation, quality standards, and timely project completion, which contributes to its growth and competitive position in the industry.

Overall, the company profile reflects that AV Engineers Pvt. Ltd. is a growing and reliable engineering organization that combines technical expertise, operational efficiency, and customer satisfaction to achieve sustainable business success in the HVAC/R sector.

CHAPTER III

INTRODUCTION TO TOPIC

Operations Exposure at AV Engineers Pvt. Ltd., Chandigarh

3.1 INTRODUCTION

Operations exposure is an essential part of understanding how a company truly functions beyond textbooks and classroom learning. While theoretical knowledge helps build a foundation, it is only through real-world exposure that one can truly grasp how organizations operate on a daily basis.

In the case of **AV Engineers Pvt. Ltd.**, this study focuses on gaining practical insight into the company's operational activities within the HVAC (Heating, Ventilation, Air Conditioning, and Refrigeration) industry. The organization deals with complex systems that require both technical expertise and efficient management. Observing how these elements come together provides a valuable learning experience.

This topic aims to give a detailed understanding of how projects are planned, executed, and managed efficiently while maintaining quality and meeting client expectations. It also highlights how coordination between different departments ensures smooth functioning.

3.2 Meaning of Operations Exposure

Operations exposure can be described as the process of learning through direct interaction with real business activities. It allows a student or trainee to move beyond theory and witness how tasks are actually performed in a professional environment.

Instead of just reading about processes, operations exposure helps in observing real-time workflows, understanding how teams collaborate, and seeing how decisions are implemented on the ground. It also provides an opportunity to identify challenges that organizations face and how they overcome them.

In simple terms, operations exposure is about **learning by observing, participating, and experiencing** the functioning of a company. It transforms theoretical knowledge into practical understanding, making learning more meaningful and impactful.

In today's fast-paced and competitive industrial environment, companies must operate efficiently to survive and grow. Organizations like **AV Engineers Pvt. Ltd.** need to carefully manage their resources, ensure timely completion of projects, and maintain high standards of service.

Every project in the HVAC industry involves multiple stages—from planning and design to installation and maintenance. Each stage requires careful coordination and decision-making. Even a small delay or mistake can impact the overall project timeline and customer satisfaction.

This study is important because it helps in understanding how such operational activities are planned, monitored, and controlled in a real-world setting. It provides insight into the challenges faced by companies and the strategies they use to maintain efficiency and quality.

3.3 Objectives Of Operation Exposure

The main objectives of studying Operations Exposure are to understand how a company manages its daily operational activities and how external factors affect business performance. In the context of AV Engineers Pvt Ltd, the objectives can be written as follows:

- **To understand the operational activities of the company-**

To study the day-to-day operations involved in HVAC&R projects, including planning, installation, maintenance, and customer service.

- **To analyze the workflow and operational process-**

To examine how different departments coordinate their work for smooth execution of projects and timely delivery of services.

- **To study resource management practices-**

To understand how the company manages manpower, machinery, materials, and technology for efficient operations.

- **To evaluate operational efficiency-**

To identify the strengths and weaknesses in operational performance and measure productivity and effectiveness.

- **To understand the impact of market and environmental factors-**

To study how changes in customer demand, technology, competition, and economic conditions influence company operations.

- **To examine quality control and safety measures-**

To analyze the standards and procedures followed by the company to maintain service quality and workplace safety.

- **To study inventory and supply chain management-**

To understand how equipment, spare parts, and materials are purchased, stored, and supplied for various projects.

- **To identify operational challenges faced by the company-**

To examine problems such as project delays, cost control, maintenance issues, and workforce management.

- **To understand customer satisfaction and service management-**

To study how the company handles customer requirements, after-sales services, and complaint resolution.

- **To suggest improvements for better operational performance-**

To provide recommendations that can help improve efficiency, reduce operational costs, and increase productivity.

- **To gain practical exposure to industrial operations-**

To develop practical knowledge and understanding of real business operations in the HVAC&R industry.

- **To bridge theoretical and practical knowledge-**

To compare academic concepts of operations management with actual industrial practices followed in the company.

3.4 Importance of Operations in the HVAC Industry

- **Helps in Understanding Business Operations**

Operations exposure gives practical knowledge about how different departments such as production, finance, purchasing, sales, and logistics work together in an organization.

- **Improves Decision Making**

By studying operations exposure, management can make better decisions related to production planning, inventory control, cost reduction, and resource utilization.

- **Identifies Operational Risks**

It helps a company identify risks like changes in raw material prices, exchange rate fluctuations, supply chain delays, and market competition that can affect business operations.

- **Increases Efficiency and Productivity**

Operations exposure helps organizations improve workflow, reduce wastage, and use resources effectively, which increases overall productivity.

- **Enhances Financial Performance**

Efficient operations reduce operational costs and improve profitability. It also helps in maintaining stable cash flow and better financial management.

- **Supports International Business Activities**

For companies dealing in imports and exports, operations exposure helps understand the impact of foreign exchange rates and international market conditions on business activities.

- **Develops Practical Knowledge**

It provides real-world exposure to students and employees by connecting theoretical concepts with practical industrial operations.

- **Improves Coordination Between Departments**

Operations exposure helps different departments communicate and coordinate effectively, leading to smooth functioning of the organization.

- **Customer Satisfaction**

Better operational management ensures timely delivery of products and services, which improves customer satisfaction and company reputation.

- **Helps in Achieving Organizational Goals**

Effective operations management supports the achievement of business objectives such as growth, profitability, quality improvement, and market expansion

3.5 Conclusion

Operations exposure plays an important role in the growth and stability of a company. It refers to the effect of changes in market conditions, exchange rates, costs, demand, and competition on the day-to-day operations and future profitability of a business. In the case of AV Engineers Pvt Ltd, operations exposure helps the company understand and manage the risks involved in purchasing materials, dealing with suppliers, handling customer requirements, and maintaining smooth business operations.

The study of operations exposure shows that effective planning, proper inventory management, cost control, quality maintenance, and timely decision-making are essential for improving organizational performance. Since AV Engineers operates in the HVAC & Refrigeration industry, fluctuations in raw material prices, technological changes, customer demand, and market competition can directly affect the company's operational efficiency and profitability.

The company can reduce operational risks by adopting better financial planning, maintaining strong relationships with suppliers and customers, using modern technology, and improving internal communication. Proper management of operations exposure also helps in achieving customer satisfaction, increasing productivity, and maintaining long-term business sustainability.

Overall, operations exposure is a significant aspect of business management because it affects both the present operations and future growth of the organization. A well-managed operational system enables AV Engineers Pvt Ltd to face market uncertainties confidently and maintain its competitive position in the industry.

CHAPTER IV

Research Methodology

Introduction

Research methodology refers to the systematic way of collecting, analyzing, and interpreting information for a particular study. It explains *how* the research has been carried out and *what methods* have been used to gather data.In this study, the research methodology focuses on understanding the operational activities of **AV Engineers Pvt. Ltd.** through practical exposure. Since the study is based on real-world observation, the approach is mainly descriptive and exploratory in nature

4.1 Review of Literature.

The review of literature is an important part of any study as it helps in understanding what previous researchers, scholars, and industry experts have already explored in a particular field. It provides a theoretical foundation and helps in identifying gaps that the current study aims to address.In the context of operations exposure in the HVAC industry, the literature focuses on concepts such as operations management, project execution, supply chain management, and service quality. These areas are highly relevant for understanding how companies like **AV Engineers Pvt. Ltd.** function in real-world conditions.

1. Concept of Operations Management

According to **William J. Stevenson**, operations management involves the planning, organizing, and supervising of processes that convert inputs into finished goods or services. He emphasizes that efficient operations are essential for achieving productivity and maintaining competitiveness.

Similarly, **Jay Heizer** and **Barry Render** highlight that operations management is the backbone of any organization, as it ensures optimal use of resources and timely delivery of

services. Their work stresses the importance of coordination between different functional areas such as production, procurement, and quality control.

These concepts are directly applicable to HVAC companies, where multiple processes must be managed simultaneously to ensure successful project execution.

2. Operations in Service and Technical Industries

Research shows that operations in service-oriented and technical industries, like HVAC, are more complex compared to manufacturing industries. This is because services are often customized based on client requirements.

According to **Richard B. Chase**, service operations require a high level of customer interaction and flexibility. He points out that the success of such industries depends on responsiveness, reliability, and service quality.

In HVAC organizations, every project is unique, requiring customized system design and installation. This aligns with the idea that operations must be adaptable and customer-focused.

3. Project Management and Execution

Project management plays a crucial role in operational efficiency. **Harold Kerzner** defines project management as the process of planning, organizing, and controlling resources to achieve specific goals within a defined timeframe.

His studies emphasize that successful project execution depends on proper planning, scheduling, and risk management. Delays, cost overruns, and poor coordination are common challenges that organizations must overcome.

This is highly relevant in the HVAC industry, where projects involve multiple stages such as design, procurement, installation, and testing. Effective project management ensures that all these stages are completed smoothly.

4. Supply Chain and Inventory Management

Efficient procurement and inventory management are essential for maintaining smooth operations. According to **Sunil Chopra** and **Peter Meindl**, supply chain management focuses on the flow of materials, information, and finances across the organization.

They highlight that proper inventory control helps in reducing costs and avoiding delays. Maintaining the right balance between demand and supply is crucial for operational efficiency.

In HVAC companies, where projects depend on timely availability of equipment and materials, effective supply chain management becomes even more important.

5. Quality Management in Operations

Quality management is another key aspect discussed in literature. **W. Edwards Deming** emphasized that quality should be built into processes rather than inspected at the end.

Similarly, **Joseph M. Juran** introduced the concept of quality planning, control, and improvement. These principles help organizations maintain consistency and meet customer expectations.

In the HVAC industry, quality is critical because system performance directly affects customer satisfaction. Proper testing, commissioning, and maintenance are necessary to ensure reliability.

6. Importance of Customer Satisfaction and After-Sales Service

Customer satisfaction has been widely discussed in service management literature. According to **Philip Kotler**, customer satisfaction is the key to long-term business success.

He highlights that after-sales services, complaint handling, and relationship management play a vital role in retaining customers. Companies that focus on customer needs and provide reliable services are more likely to succeed.

In HVAC businesses, maintenance services and annual contracts are essential for building long-term relationships with clients.

4.2 Need Of Study

- To understand the concept of operations exposure in the company.
- To identify the factors affecting operational efficiency and business performance.
- To study how operational risks influence cost, productivity, and profitability.
- To analyze the existing operational practices followed by the company.
- To examine the impact of market fluctuations on company operations.
- To evaluate the effectiveness of operational planning and control systems.
- To identify challenges faced by the company in managing operations exposure.
- To collect primary data from employees and management through questionnaires/interviews.
- To provide practical suggestions for improving operational performance and reducing exposure.
- To help the researcher gain practical knowledge about industrial operations and management.
- To bridge the gap between theoretical concepts and real business practices.
- To support future decision-making and strategic planning in the organization.

4.3 Scope Of The Study

- To study the concept and importance of operations exposure in a business organization.
- To analyze the impact of operational risks on the company’s performance and profitability.
- To examine the effect of fluctuations in costs, supply, production, and market conditions on business operations.

- To understand how the company manages operational challenges in domestic and international markets.
- To evaluate the role of operational planning and control in reducing exposure and uncertainty.
- To study the operational procedures followed by the company in manufacturing and project execution.
- To identify the major factors affecting operational efficiency and productivity.
- To analyze the strategies adopted by the company for risk management and smooth operations.
- To collect and interpret employee and management responses through questionnaires/interviews.
- To suggest measures for improving operational performance and minimizing operational exposure.
- The study is limited to the operational activities and practices of the company during the research period.
- The findings of the study may help in improving future operational decision-making and business growth.

4.4 Objectives

- To understand the operational structure
- To analyse the workflow of HVAC&R projects
- To study inventory and material management practices
- To identify challenges faced in daily operations
- To evaluate cost control and budgeting in operation

4.5 Research Design

❖ **Type of Research**

1. Descriptive Research

The study mainly uses descriptive research because it describes and analyzes the operational exposure faced by the company. It helps in understanding:

➢ Nature of operational exposure

- ➢ Effect on company operations
- ➢ Risk management practices
- ➢ Employee and management opinions

2. Analytical Research

Analytical research is also used because the collected data is carefully analyzed to identify the causes and effects of operational exposure on the organisation

4.6 Data source

a. Primary Data

Primary data refers to the information collected directly from the source. In this study, it has been collected through:

- **Direct Observation:**
  Observing day-to-day operations such as project execution, installation, and coordination between departments.
- **Interaction with Employees:**
  Informal discussions with engineers, technicians, and staff members to understand their roles and responsibilities.
- **On-site Exposure:**
  Visiting project sites and observing how HVAC systems are installed, tested, and maintained.

This method helped in gaining first-hand knowledge of how operations are carried out in real situations.

❖ **Hypothesis**

Null Hypothesis (H_0)

There is no significant impact of operations exposure on the financial and operational performance of AV Engineers Pvt Ltd.

Alternative Hypothesis (H_1)

There is a significant impact of operations exposure on the financial and operational performance of AV Engineers Pvt Ltd.

4.7 Sample Design

The study includes employees, managers, finance staff, and operational staff of the company who are directly or indirectly related to operational activities and financial decision-making.

- **Sample Unit**

The sample unit consists of selected employees and management personnel of AV Engineers Pvt Ltd.

- **Sample Size**

A sample of 30–50 respondents may be selected for the study, depending upon availability and convenience.

- **Sampling Technique**

The study uses Convenience Sampling Method because respondents are selected based on accessibility and willingness to provide information.

4.8 Tools & Analysis

The collected data for the study can be analyzed with the help of the following tools:

1. Percentage Analysis

Percentage method is used to analyze responses collected from employees and management through questionnaires. It helps in understanding the proportion of responses for different questions.

2. Tabular Analysis

Data collected from respondents is presented in the form of tables for easy understanding and comparison.

Helps in systematic presentation of data

Makes interpretation simple and clear

4.9 Limitation Of Study

1. Limited Time Duration

The biggest limitation was time. The duration of the training or study was not long enough to understand every aspect of the organization in depth.Operations in an HVAC company involve many stages like planning, design, procurement, execution, and maintenance. Due to limited time, it was not possible to observe each stage completely or multiple times. Some processes were only seen partially, which may affect the depth of understanding.

2. Limited Access to Information

Not all information within the company was accessible. Some data related to projects, costs, clients, or internal strategies is confidential.Because of this, the study mainly relied on general information, observation, and discussions rather than detailed internal records. This limitation made it difficult to analyze certain aspects like exact costing, profit margins, or strategic decisions.

3. Study Limited to One Company

The study is focused only on **AV Engineers Pvt. Ltd.**, which means the findings are based on the working style of a single organization.Different companies may follow different processes, technologies, or management styles. So, the results of this study cannot be fully generalized to the entire HVAC industry.

4. Lack of Practical Involvement

Most of the learning was based on observation rather than direct involvement in decision-making or execution.As a trainee or student, there were certain limitations in participating actively in critical tasks. This means the understanding is based on what was observed and explained, not on full hands-on responsibility.

5. Dependence on Employee Interaction

A part of the study depended on discussions with employees. Sometimes, employees may be busy or may not be able to explain everything in detail. Also, the information shared may vary depending on the person's experience or perspective, which can affect the accuracy or consistency of the data collected.

6. Limited Exposure to All Departments

Due to time and access constraints, it was not possible to explore every department in detail. Some areas like advanced design processes, financial planning, or higher-level management decisions were only understood at a basic level. This limits the overall depth of the study.

CHAPTER V

DATA ANALYSIS & INTERPRETATION

Introduction

Data analysis and interpretation is an important part of the research process. It helps in understanding the opinions and responses collected through questionnaires. The data for this study was collected through a structured questionnaire from respondents associated with HVAC project operations.

Question 1. Age group of the respondent ?

Table 5.1

AGE GROUP	PERCENTAGE
20-25	80%
25-30	20%

Interpretation

The above table shows that the majority of respondents (80%) belong to the age group of 20-25 years, while 20% belong to the age group of 25-30 years. This indicates that young professionals are actively involved in project operations.

Question 2. Company preparation for project plan before execution ?

Table 5.2

yes	80%
no	20%

Interpretation

The data indicates that 80% of respondents agreed that the company prepares a project plan before execution, while 20% stated otherwise. This shows that project planning is an important operational practice in the company.

Question 3. Proper schedule preparation for project completion ?

Table 5.3

Always	Sometime	Rarely
50%	30%	20%

Interpretation

The table reveals that 50% of respondents believe proper schedules are always prepared for project completion, whereas 30% said sometimes and 20% said rarely. This indicates that scheduling practices exist but require further improvement.

Question 4. Which stage consumes the most time in HVAC projects ?

Table 5.4

Procurement	Installation	Design	Testing & Commisioning
40%	30%	20%	10%

Interpretation

The analysis shows that procurement consumes the maximum time in HVAC projects according to 40% of respondents. Installation is the second most time-consuming stage. Delays in procurement directly affect project timelines.

Question 5. What are the major causes of project delay?

Table 5.5

Material Supply Delay	Financial Issue	Client Side Delay	Labour Issues
50%	20%	20%	10%

Interpretation

The majority of respondents (50%) identified material supply delay as the major cause of project delay. Financial issues and client-side delays were also important factors affecting project operations.

Question 6. Is the cost estimate prepared before starting the project?

Table 5.6

Always	Sometime
60%	40%

Interpretation

The above table shows that 60% of respondents stated that cost estimates are always prepared before starting the project, while 40% said sometimes. This indicates that cost estimation is generally practiced within the company.

Question 7. Does the company maintain inventory stock ?

Table 5.7

Yes	No
90%	10%

Interpretation

The analysis reveals that 90% of respondents agreed that the company maintains inventory stock. Proper inventory management helps reduce operational disruptions and delays.

Question 8. Does delay in material supply affect project profitability?

Table 5.8

Yes	No
80%	20%

Interpretation

The majority of respondents (80%) believe that delay in material supply affects project profitability. This indicates the importance of timely procurement and supply chain management.

Question 9. Does delay in client payment affect operations ?

Table 5.9

Yes	No
80%	20%

Interpretation

The table indicates that delayed client payments negatively affect company operations according to 80% of respondents. Proper payment management is therefore essential for smooth operations.

Question 10. Do you think digital tools can improve efficiency ?

Table 5.10

Yes	No
60%	40%

Interpretation

All respondents agreed that digital tools can improve operational efficiency. This highlights the growing importance of technology in project management and operational control.

Chapter VI

FINDING, CONCLUSION & SUGGESTION

Findings -

- Most respondents belong to the young workforce category of 20-25 years.
- The company generally prepares project plans before execution.
- Procurement is the most time-consuming stage in HVAC projects.
- Material supply delay is the major reason for project delays.
- Cost estimation practices are followed in the company.
- Inventory stock is maintained properly.
- Delay in material supply and client payment significantly affects operations and profitability.
- Digital tools are considered highly beneficial for improving efficiency.

Conclusion-

The study on Operations Exposure at AV Engineers Pvt Ltd highlights the importance of effective operational management in achieving successful project execution and maintaining profitability in the HVAC industry. The analysis of the responses collected through the questionnaire indicates that operational exposure directly influences the company's productivity, cost efficiency, project completion time, and customer satisfaction.The findings reveal that the company generally follows proper project planning and scheduling practices before the execution of projects. Most respondents agreed that project plans and cost estimates are prepared in advance, which helps the organization reduce uncertainties and improve operational efficiency. Proper planning also supports better utilization of resources, labour, machinery, and inventory.The study further shows that procurement and material supply are the most critical operational areas affecting project performance. Delays in material procurement and supply were identified as the major reasons for project delays. Such delays increase project costs, affect profitability, and reduce overall operational efficiency. Therefore, effective supplier management and inventory control are essential for minimizing operational

risks.Another important finding of the study is that delayed client payments negatively impact the company's day-to-day operations. Cash flow interruptions caused by late payments create financial pressure on the organization and affect smooth project execution. This emphasizes the need for strong financial planning and efficient payment collection systems.The research also indicates that inventory management plays a vital role in reducing operational exposure. Most respondents confirmed that maintaining inventory stock helps avoid unnecessary delays and ensures uninterrupted workflow. Proper inventory management improves project coordination and reduces dependency on emergency procurement.In addition, all respondents agreed that digital tools and modern technologies can improve operational efficiency. The use of project management software, inventory tracking systems, and digital communication tools can help the company monitor project progress, reduce errors, improve coordination, and ensure timely completion of projects. Overall, the study concludes that operational exposure significantly affects the performance and profitability of HVAC projects at AV Engineers Pvt Ltd. Effective project planning, timely procurement, proper inventory management, strong financial control, and adoption of digital technologies are necessary to reduce operational risks and improve organizational performance. By implementing these measures, the company can enhance efficiency, reduce delays, improve customer satisfaction, and achieve long-term business growth.

Suggestion-

- Improve procurement planning & supplier coordination.
- Strengthen inventory management system.
- Adopt advanced digital project management tools.
- Improve payment procedure from client.
- Prepare a detailed project scheduled for timely completion.
- Conduct regular operational monitoring to reduce delays.

BIBLIOGRAPHY

Kothari, C.R. (2019). *Research Methodology: Methods and Techniques*. New Age International Publishers.

This book is one of the most widely used references for understanding research methodology.

It provided a strong foundation for designing the research framework of this study. Concepts such as research design, data collection methods, sampling techniques, and limitations of research were understood with the help of this source. It played an important role in structuring the methodology section of the project.

Sharma, R., & Verma, P. (2021). *Study on HVAC Project Execution Challenges*. International Journal of Engineering Management.

This research paper focuses on the various challenges faced during HVAC project execution, such as delays, coordination issues, and cost management. It helped in understanding real-world problems in the HVAC industry and supported the analysis of operational challenges observed at **AV Engineers Pvt. Ltd.**.

Gupta, S. (2020). *Maintenance Management in HVAC Systems*. Journal of Facility Management.

This article explains the importance of regular maintenance and servicing in HVAC systems. It highlights the role of preventive maintenance, troubleshooting, and performance monitoring. This source was useful in understanding the after-sales services and maintenance practices followed in the company.

Singh, A., Kumar, R., & Mehta, P. (2022). *Supply Chain Efficiency in Engineering Services*. Indian Journal of Operations Management.

This paper discusses how efficient supply chain and inventory management contribute to successful project execution. It provided insights into procurement practices, vendor management, and inventory control, which are essential parts of operations in HVAC companies.

Patel, M. (2023). *Digital Transformation in HVAC Operations*. Construction Technology Review.

This study highlights the growing role of digital tools and technologies in improving HVAC

operations. It explains how automation, data tracking, and smart systems can enhance efficiency. This helped in understanding modern trends in the HVAC industry.

Company Internal Records of AV Engineers Pvt. Ltd..

The internal documents, brochures, and reports of the company were an important source of practical information. These records provided insights into the company's projects, operational processes, and service offerings. They helped in understanding how theoretical concepts are applied in real-life situations.

Interaction with Employees of AV Engineers Pvt. Ltd..

Informal discussions with engineers, technicians, and staff members played a key role in collecting primary data. These interactions helped in understanding daily operations, challenges faced during project execution, and the coordination between different departments.

On-site Observations during Training at AV Engineers Pvt. Ltd..

Direct observation of project sites provided first-hand knowledge of installation processes, testing procedures, and maintenance activities. This practical exposure helped in connecting theoretical knowledge with real-world practices.

www.trade.gov

This website provided useful information about industry standards, trade practices, and global perspectives related to engineering services and HVAC systems. It helped in understanding the broader industrial environment.

www.energy.gov

This website offered valuable insights into energy efficiency, HVAC technologies, and environmental considerations. It helped in understanding the importance of energy-saving practices in HVAC operations.

www.google.com

Google was used as a general search tool to find relevant articles, definitions, and supporting materials related to operations management and HVAC systems.

www.wikipedia.org

Wikipedia was used for basic understanding and quick reference of concepts related to HVAC systems, operations management, and related topics. The information was cross-checked with other reliable sources.

Appendices

Questionnaire-

Email*

Name*

What is your age group *

- ☐ 20-25
- ☐ 25-30
- ☐ 30-35

Does the company prepare a project plan before starting executive? *

- ☐ Yes
- ☐ No

Is a proper schedule prepared for project completion *

- ☐ Always
- ☐ Sometimes
- ☐ Rarely

Which stage consumes the most time in HVAC projects *

- [ ] Design
- [ ] Procurement
- [ ] Installation
- [ ] Testing and commissioning

What are the major causes of project delay? *

- [ ] Material supply delay
- [ ] Client side delay
- [ ] Labor shortage
- [ ] Financial issue

Are the cost estimates prepared before starting the project? *

- [ ] Always
- [ ] Sometimes
- [ ] Rarely

Does the company maintain inventory stock? *

- [ ] Yes
- [ ] No

Does delay in material supply affect project profitability? *

- [ ] Yes
- [ ] No

Does delay in client payment affect operations? *

- [ ] Yes
- [ ] No

Do you think digital tools can improve efficiency? *

- [ ] Yes
- [ ] No